

Artificial intelligence (AI) is the capability of computational systems to perform tasks that are typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. AI has been used in applications throughout industry and academia. Within the field of artificial intelligence, there are multiple subfields. The subfield of machine learning has been used for various scientific and commercial purposes, including language translation, image recognition, decision-making, credit scoring, and e-commerce. Since the 2020s, massive advancements have been made in the field of generative artificial intelligence (GenAI), which uses generative models to create text, images, music, videos, and other forms of data. This article describes applications of AI in different sectors.

Agriculture

In agriculture, AI has been proposed as a way for farmers to identify areas that need irrigation, fertilization, or pesticide treatments to increase yields, thereby improving efficiency. AI has been used to attempt to classify livestock pig call emotions, automate greenhouses, detect diseases and pests, and optimize irrigation.

AI-assisted software development

Architecture and design

Business

A 2023 study found that generative AI increased productivity by 15% in contact centers. Another 2023 study found it increased productivity by up to 40% for writing tasks. An August 2025 review by MIT found that of surveyed companies, 95% did not report any improvement in revenue from the use of AI. In September 2025, a Harvard Business Review article described how increased use of AI does not automatically lead to increases in revenue or actual productivity. Referring to "AI generated work content that masquerades as good work, but lacks the substance to meaningfully advance a given task", the article coined the term

workslop. Per studies done in collaboration with the Stanford Social Media Lab, workslop does not improve productivity and undermines trust and collaboration among colleagues.

In telehealth, agentic AI is reportedly facilitating the creation of large business models (millions in annual profit) with 1–2 employees, such as MEDVi, which as of August 2025 only had 2 employees and ~\$75 million in annual profit for GLP-1 weight-loss telehealth services.

Chatbots

Computer science

Programming assistance

AI-assisted software development

AI can be used for real-time code completion, chat, and automated test generation. These tools are typically integrated with editors and IDEs as plugins. AI-assisted software development systems differ in functionality, quality, speed, and approach to privacy. Creating software primarily via AI is known as "vibe coding". Code created or suggested by AI can be incorrect or inefficient. The use of AI-assisted coding can potentially speed-up software development, but it can also slow-down the process by creating more work when debugging and testing. The rush to prematurely adopt AI technology can also incur additional technical debt. AI also requires additional consideration and careful review for cybersecurity, since AI coding software is trained on a wide range of code of inconsistent quality and often replicates poor practices.

Neural network design

AI can be used to create other AIs. For example, around November 2017, Google's AutoML project to evolve new neural net topologies created NASNet, a system optimized for ImageNet and POCO F1. NASNet's performance exceeded all previously published performance on ImageNet.

Quantum computing

Research and development of quantum computers has been performed with machine learning algorithms. For example, there is a prototype, photonic, quantum memristive device for neuromorphic computers (NC)/artificial neural networks and NC-using quantum materials with some variety of potential neuromorphic computing-related applications. The use of quantum machine learning for quantum simulators has been proposed for solving physics and chemistry problems.

Historical contributions

AI researchers have created many tools to solve difficult problems in computer science. Many of their inventions have been adopted by mainstream computer science and are no longer considered AI. The following were originally developed in AI laboratories:

Time sharing

Interactive interpreters

Graphical user interfaces and the computer mouse

Rapid application development environments

The linked list data structure

Automatic storage management

Symbolic programming

Functional programming

Dynamic programming

Object-oriented programming

Optical character recognition

Constraint satisfaction

Customer service

Human resources

AI programs have been used in hiring processes to screen resumes and rank candidates based on their qualifications, predict a candidate's likelihood of success in a given role, and automate repetitive communication tasks using chatbots. Studies on these programs have identified tendencies for gender bias, favoring male names and male-coded characteristics, as well as bias against disabled candidates and racial minorities.

Online and telephone customer service

AI underlies avatars (automated online assistants) on web pages. It can reduce operation and training costs. In 2016, Pypestream automated customer service for its mobile application to streamline communication with customers.

A Google app analyzes language and converts speech into text. The platform can identify angry customers through their language and respond appropriately. Amazon uses a chatbot for customer service that can perform tasks like checking the status of an order, cancelling orders, offering refunds, and connecting the customer with a human representative.

Hospitality

In the hospitality industry, AI is used to reduce repetitive tasks, analyze trends, interact with guests, and predict customer needs. AI hotel services come in the form of a chatbot, application, virtual voice assistant, and service robots.

Education

In educational institutions, AI has been used to automate routine tasks such as attendance tracking, grading, and marking. AI tools have also been used to monitor student progress and

analyze learning behaviors, with the goal of facilitating timely interventions for students facing academic challenges.

As of 2025 and 2026, respectively, about 80% of high school students and nearly 60% of college students were also using AI. Some schools and universities proposed restrictions on students' usage due to issues with increased cheating, over-reliance on AI tools, and widening of digital inequities.

Energy and environment

Energy system

The U.S. Department of Energy wrote in an April 2024 report that AI may have applications in modeling power grids, reviewing federal permits with large language models (LLMs), predicting levels of renewable energy production, and improving the planning process for electrical vehicle charging networks. Other studies have suggested that machine learning can be used for energy consumption prediction and scheduling (e.g., to help with renewable energy intermittency management).

Environmental monitoring

Autonomous ships that monitor the ocean, AI-driven satellite data analysis, passive acoustics or remote sensing and other applications of environmental monitoring make use of machine learning.

For example, "Global Plastic Watch" is an AI-based satellite-monitoring platform for analysis/tracking of plastic waste sites to help with the prevention of plastic pollution—primarily ocean pollution—by helping identify who mismanages plastic waste and where dumping into oceans occurs.

Early-warning systems

Machine learning can be used to spot early warning signs of disasters and environmental issues, possibly including natural pandemics, earthquakes, landslides, heavy rainfall, long-term water supply vulnerability, tipping-points of ecosystem collapse, cyanobacterial bloom outbreaks, and droughts.

Economic and social challenges

In 2023, the University of Southern California launched the Center for Artificial Intelligence in Society, with the goal of using AI to address problems such as homelessness. Stanford University researchers use AI to analyze satellite images to identify high poverty areas.

Entertainment and media

Media

AI applications analyze media content such as movies, TV programs, advertisement videos, and user-generated content. The solutions often involve computer vision.

Typical scenarios include the analysis of images using object recognition or facial recognition techniques, or the analysis of video for recognizing scenes, objects, or faces. AI-based media analysis can facilitate media search, the creation of descriptive keywords for content, content policy monitoring (such as verifying the suitability of content for a particular TV viewing time), speech to text for archival or other purposes, and the detection of logos, products, or celebrity faces for ad placement.

Motion interpolation

Pixel-art scaling algorithms

Image scaling

Image restoration

Photo colorization

Film restoration and video upscaling

Photo tagging

Text-to-image models (such as DALL-E, Midjourney, and Stable Diffusion)

Image to video

Text to video (such as Make-A-Video from Meta, Imagen video and Phenaki from Google)

Text to music with AI models (such as MusicLM)

Text to speech (such as ElevenLabs and 15.ai)

Motion capture

Deep-fakes

Deep-fakes can be used for comedic purposes but are better known for fake news and hoaxes.

Deepfakes can portray individuals in harmful or compromising situations, causing significant reputational damage and emotional distress—especially when the content is defamatory or violates personal ethics. While defamation and false light laws offer some recourse, their focus on false statements rather than fabricated images or videos often leaves victims with limited legal protection and a challenging burden of proof. Audio deepfakes and AI software capable of detecting deep-fakes and cloning human voices have also been developed.

In January 2016, the Horizon 2020 program financed the InVID Project to help journalists and researchers detect fake documents, made available as browser plugins.

In September 2018, U.S. Senator Mark Warner proposed penalizing social media companies that allow sharing of deep-fake documents on their platforms. In 2018, Darius Afchar and Vincent Nozick found a way to detect faked content by analyzing the mesoscopic properties of video frames. DARPA gave \$68 million to work on deep-fake detection.

Video surveillance analysis and manipulated media detection

AI algorithms have been used to detect deepfake videos.

Video production

Artificial intelligence is also starting to be used in video production, with tools and software being developed that utilize generative AI in order to create new video, or alter existing video. Some of the major tools that are being used in these processes currently are DALL-E, Mid-journey, and Runway. Way mark Studios utilized the tools offered by both DALL-E and Mid-journey to create a fully AI generated film called The Frost in the summer of 2023. Way mark Studios is experimenting with using these AI tools to generate advertisements and commercials for companies in mere seconds. Yves Bergquist, a director of the AI & Neuroscience in Media Project at USC's Entertainment Technology Center, says post production crews in Hollywood are already using generative AI, and predicts that in the future more companies will embrace this new technology.

Music

AI has been used to compose music of various genres.

David Cope created an AI called Emily Howell that managed to become well known in the field of algorithmic computer music. The algorithm behind Emily Howell is registered as a US patent.

In 2012, AI Iamus created the first complete classical album.

AIVA (Artificial Intelligence Virtual Artist), composes symphonic music, mainly classical music for film scores. It achieved a world first by becoming the first virtual composer to be recognized by a musical professional association.

Melomics creates computer-generated music for stress and pain relief.

The Watson Beat uses reinforcement learning and deep belief networks to compose music on a simple seed input melody and a select style. The software was open sourced and musicians such as Taryn Southern collaborated with the project to create music.

South Korean singer, Hayeon's, debut song, "Eyes on You" was composed using AI which was supervised by real composers, including NUVO.

Writing and reporting

Narrative Science sells computer-generated news and reports. It summarizes sporting events based on statistical data from the game. It also creates financial reports and real estate analyses. Automated Insights generates personalized recaps and previews for Yahoo Sports Fantasy Football.

Yseop, uses AI to turn structured data into natural language comments and recommendations. Yseop writes financial reports, executive summaries, personalized sales or marketing documents and more in multiple languages, including English, Spanish, French, and German.

TALESPIN made up stories similar to the fables of Aesop. The program started with a set of characters who wanted to achieve certain goals. Mark Riedl and Vadim Bulitko asserted that the essence of storytelling was experience management, or "how to balance the need for a coherent story progression with user agency, which is often at odds".

While AI storytelling focuses on story generation (character and plot), story communication also received attention. In 2002, researchers developed an architectural framework for narrative prose generation. They faithfully reproduced text variety and complexity on stories such as Little Red Riding Hood. In 2016, a Japanese AI co-wrote a short story and almost won a literary prize.

South Korean company Hanteo Global uses a journalism bot to write articles.

Literary authors are also exploring uses of AI. An example is David Jhave Johnston's work *ReRites* (2017–2019), where the poet created a daily rite of editing the poetic output of a neural network to create a series of performances and publications.

Sports writing

In 2010, artificial intelligence used baseball statistics to automatically generate news articles. This was launched by The Big Ten Network using software from Narrative Science.

After being unable to cover every Minor League Baseball game with a large team, Associated Press collaborated with Automated Insights in 2016 to create game recaps that were automated by artificial intelligence.

UOL in Brazil expanded the use of AI in its writing. Rather than just generating news stories, they programmed the AI to include commonly searched words on Google.

El Pais, a Spanish news site that covers many things including sports, allows users to make comments on each news article. They use the Perspective API to moderate these comments and if the software deems a comment to contain toxic language, the commenter must modify it in order to publish it.

A local Dutch media group used AI to create automatic coverage of amateur soccer, set to cover 60,000 games in just a single season. NDC partnered with United Robots to create this algorithm and cover what would have never been possible before without an extremely large team.

Lede AI has been used in 2023 to take scores from high school football games to generate stories automatically for the local newspaper. This was met with significant criticism from readers for the very robotic diction that was published. With some descriptions of games being a "close encounter of the athletic kind," readers were not pleased and let the publishing company, Gannett, know on social media. Gannett has since halted their use of Lede AI until they come up with a solution for what they call an experiment.

Wikipedia

Millions of its articles have been edited by bots which however are usually not artificial intelligence software. Many AI platforms use Wikipedia data, mainly for training machine

learning applications. There is research and development of various artificial intelligence applications for Wikipedia such as for identifying outdated sentences, detecting covert vandalism or recommending articles and tasks to new editors.

Machine translation (see above) has also be used for translating Wikipedia articles and could play a larger role in creating, updating, expanding, and generally improving articles in the future. A content translation tool allows editors of some Wikipedias to more easily translate articles across several select languages.

Video games

In video games, AI is routinely used to generate behavior in non-player characters (NPCs). In addition, AI is used for pathfinding. Games with less typical AI include the AI director of Left 4 Dead (2008) and the neuroevolutionary training of platoons in Supreme Commander 2 (2010). AI is also used in Alien Isolation (2014) as a way to control the actions the Alien will perform next.

Games have been a major application of AI's capabilities since the 1950s. In the 21st century, AIs have beaten human players in many games, including chess (Deep Blue), Jeopardy! (Watson), Go (AlphaGo), poker (Pluribus and Cepheus), E-sports (StarCraft), and general game playing (AlphaZero and MuZero).

Kuki AI is a set of chatbots and other apps which were designed for entertainment and as a marketing tool.

Visual images

The first AI art program, called AARON, was developed by Harold Cohen in 1968 with the goal of being able to code the act of drawing. It started by creating simple black and white drawings, and later to painting using special brushes and dyes that were chosen by the program itself without mediation from Cohen.

AI platforms such as DALL-E, Stable Diffusion, Imagen, and Midjourney have been used for generating visual images from inputs such as text or other images. Some AI tools allow users to input images and output changed versions of that image, such as to display an object or product in different environments. AI image models can also attempt to replicate the specific styles of artists, and can add visual complexity to rough sketches.

AI has been used to generate quantitative analysis of existing digital art collections. Two computational methods, close reading and distant viewing, are the typical approaches used to analyze digitized art. While distant viewing includes the analysis of large collections, close reading involves one piece of artwork.

Computer animation

In 2023, Netflix of Japan's usage of AI to generate background images for short *The Dog & the Boy* was met with backlash online.

Finance

Financial institutions have long used artificial neural network systems to detect charges or claims outside of the norm, flagging these for human investigation. The use of AI in banking began in 1987 when Security Pacific National Bank launched a fraud prevention task-force to counter the unauthorized use of debit cards.

Banks use AI to organize operations for bookkeeping, investing in stocks, and managing properties. AI can adapt to changes during non-business hours. AI is used to combat fraud and financial crimes by monitoring behavioral patterns for any abnormal changes or anomalies.

The use of AI in applications such as online trading and decision-making has changed major economic theories. For example, AI-based buying and selling platforms estimate personalized demand and supply curves, thus enabling individualized pricing. AI systems reduce

information asymmetry in the market and thus make markets more efficient. The application of artificial intelligence in the financial industry can alleviate the financing constraints of non-state-owned enterprises, especially for smaller and more innovative enterprises.

Trading and investment

Large financial institutions use AI to assist with their investment practices. BlackRock's AI engine, Aladdin, is used both within the company and by clients to help with investment decisions. Its functions include the use of natural language processing to analyze text such as news, broker reports, and social media feeds. It then gauges the sentiment on the companies mentioned and assigns a score. Banks such as UBS and Deutsche Bank use SQREEM (Sequential Quantum Reduction and Extraction Model) to mine data to develop consumer profiles and match them with wealth management products.

Underwriting

Online lender Upstart uses machine learning for underwriting.

ZestFinance's Zest Automated Machine Learning (ZAML) platform is used for credit underwriting. This platform uses machine learning to analyze data, including purchase transactions and how a customer fills out a form, to score borrowers. The platform is handy for assigning credit scores to those with limited credit histories.

Audit

AI makes continuous auditing possible. Potential benefits include reducing audit risk, increasing the level of assurance, and reducing audit duration.

Continuous auditing with AI allows real-time monitoring and reporting of financial activities and provides businesses with timely insights that can lead to quick decision-making.

Anti-money laundering

AI software, such as LaundroGraph which uses contemporary suboptimal datasets, could be used for anti-money laundering (AML).

History

In the 1980s, AI started to become prominent in finance as expert systems were commercialized. For example, Dupont created 100 expert systems, which helped them to save almost \$10 million per year. One of the first systems was the Pro-trader expert system that predicted the 87-point drop in the Dow Jones Industrial Average in 1986. "The major junctions of the system were to monitor premiums in the market, determine the optimum investment strategy, execute transactions when appropriate and modify the knowledge base through a learning mechanism."

One of the first expert systems to help with financial plans was PlanPowerm and Client Profiling System, created by Applied Expert Systems (APEX). It was launched in 1986. It helped create personal financial plans for people.

In the 1990s, AI was applied to fraud detection. In 1993, FinCEN Artificial Intelligence System (FAIS) was launched. It was able to review over 200,000 transactions per week, and over two years, it helped identify 400 potential cases of money laundering equal to \$1 billion. These expert systems were later replaced by machine learning systems.

Outside finance, the late 1980s and early 1990s also saw expert systems used in technical and environmental domains. For example, researchers built a fishway design advisor to recommend fish passage structures under varying hydraulic and biological conditions using the VP-Expert shell. Transportation researchers applied the same shell to balance airport capacity with noise-mitigation plans. In agriculture, a potato insect expert system (PIES) supported pest management decisions for Colorado potato beetle. The U.S. Environmental Protection Agency's CORMIX system for modeling pollutant discharges combined rules with

Fortran hydrodynamic models.

Regulatory developments in the EU

In the European Union, the Artificial Intelligence Act (Regulation (EU) 2024/1689) classifies several finance-sector uses of AI as "high-risk", including systems used to evaluate the creditworthiness of natural persons or to establish a credit score and AI used for risk assessment and pricing in life or health insurance. These systems must meet requirements on risk management, data governance, technical documentation and logging, transparency, and human oversight. The Act's obligations are phased in: prohibitions and AI-literacy rules apply from 2 February 2025, governance and most GPAI duties from 2 August 2025, the bulk of obligations from 2 August 2026, and certain safety-component high-risk obligations from 2 August 2027.

Health

Healthcare

AI in healthcare is often used for classification, to evaluate a CT scan or electrocardiogram or to identify high-risk patients for population health. AI is helping with the high-cost problem of dosing. One study suggested that AI could save \$16 billion. In 2016, a study reported that an AI-derived formula derived the proper dose of immunosuppressant drugs to give to transplant patients. Current research has indicated that non-cardiac vascular illnesses are also being treated with artificial intelligence (AI). For certain disorders, AI algorithms can aid in diagnosis, recommended treatments, outcome prediction, and patient progress tracking. As AI technology advances, it is anticipated that it will become more significant in the healthcare industry.

The early detection of diseases like cancer is made possible by AI algorithms, which diagnose diseases by analyzing complex sets of medical data. For example, the IBM Watson system

might be used to comb through massive data such as medical records and clinical trials to help diagnose a problem. Microsoft's AI project Hanover helps doctors choose cancer treatments from among the more than 800 medicines and vaccines. Its goal is to memorize all the relevant papers to predict which (combinations of) drugs will be most effective for each patient. Myeloid leukemia is one target. Another study reported on an AI that was as good as doctors in identifying skin cancers. Another project monitors multiple high-risk patients by asking each patient questions based on data acquired from doctor/patient interactions. In one study done with transfer learning, an AI diagnosed eye conditions similar to an ophthalmologist and recommended treatment referrals.

Another study demonstrated surgery with an autonomous robot. The team supervised the robot while it performed soft-tissue surgery, stitching together a pig's bowel judged better than a surgeon.

Artificial neural networks are used as clinical decision support systems for medical diagnosis, such as in concept processing technology in EMR software.

Other healthcare tasks thought suitable for an AI that are in development include:

Screening

Companion robots for elder care

Drug creation (e.g. by identifying candidate drugs and by using existing drug screening data such as in life extension research)

Clinical training

Identifying genomic pathogen signatures of novel pathogens or identifying pathogens via physics-based fingerprints (including pandemic pathogens)

Helping link genes to their functions, otherwise analyzing genes and identification of novel biological targets

Help development of biomarkers

Help tailor therapies to individuals in personalized medicine/precision medicine

Workplace health and safety

AI-enabled chatbots decrease the need for humans to perform basic call center tasks, and machine learning in sentiment analysis can spot fatigue in order to prevent overwork.

Decision support systems can potentially prevent industrial disasters and make disaster response more efficient. For manual workers in material handling, predictive analytics has been proposed to reduce musculoskeletal injury.

AI can attempt to process workers' compensation claims. AI has been proposed for detection of accident near misses, which are underreported.

Biochemistry

Machine learning has been used for drug design, drug discovery and development, drug repurposing, improving pharmaceutical productivity, and clinical trials.

Computer-planned syntheses via computational reaction networks, described as a platform that combines "computational synthesis with AI algorithms to predict molecular properties", has been used in drug-syntheses, and developing routes for recycling 200 industrial waste chemicals into important drugs and agrochemicals (chemical synthesis design). It has also been used to explore the origins of life on Earth.

Deep learning has been used with databases for the development of a 46-day process to design, synthesize and test a drug which inhibits enzymes of a particular gene, DDR1. DDR1 is involved in cancers and fibrosis which is one reason for the high-quality datasets that enabled these results.

The AI program AlphaFold 2 can determine the 3D structure of a (folded) protein in hours rather than the months required by earlier automated approaches and was used to provide the likely structures of all proteins in the human body and essentially all proteins known to

science (more than 200 million).

Language processing

Language translation

Speech translation technology attempts to convert one language's spoken words into another language. This potentially reduces language barriers in global commerce and cross-cultural exchange, enabling speakers of various languages to communicate with one another.

AI has been used to automatically translate spoken language and textual content in products such as Microsoft Translator, Google Translate, and DeepL Translator. Additionally, research and development are in progress to decode and conduct animal communication.

Meaning is conveyed not only by text, but also through usage and context (see semantics and pragmatics). As a result, the two primary categorization approaches for machine translations are statistical machine translation (SMT) and neural machine translations (NMTs). The old method of performing translation was to use statistical methodology to forecast the best probable output with specific algorithms. However, with NMT, the approach employs dynamic algorithms to achieve better translations based on context.

Law and government

Government

AI facial recognition systems are used for mass surveillance, notably in China. In 2019, Bengaluru, India deployed AI-managed traffic signals. This system uses cameras to monitor traffic density and adjust signal timing based on the interval needed to clear traffic.

Law

Legal analysis

AI is a mainstay of law-related professions. Algorithms and machine learning do some tasks

previously done by entry-level lawyers. While its use is common, it is not expected to replace most work done by lawyers in the near future.

The electronic discovery industry uses machine learning to reduce manual searching.

Law enforcement and legal proceedings

Law enforcement has begun using facial recognition systems (FRS) to identify suspects from visual data. FRS results have proven to be more accurate when compared to eyewitness results. Furthermore, FRS has shown to have much a better ability to identify individuals when video clarity and visibility are low in comparison to human participants.

COMPAS is a commercial system used by U.S. courts to assess the likelihood of recidivism.

One concern relates to algorithmic bias, AI programs may become biased after processing data that exhibits bias. ProPublica claims that the average COMPAS-assigned recidivism risk level of black defendants is significantly higher than that of white defendants.

In 2019, the city of Hangzhou, China established a pilot program artificial intelligence-based Internet Court to adjudicate disputes related to ecommerce and internet-related intellectual property claims. Parties appear before the court via videoconference and AI evaluates the evidence presented and applies relevant legal standards.

Manufacturing

Sensors

Artificial intelligence has been combined with digital spectrometry by IdeaCuria Inc., enable applications such as at-home water quality monitoring.

Toys and games

In the 1990s, early artificial intelligence tools controlled Tamagotchis and Giga Pets, the Internet, and the first widely released robot, Furby. Aibo was a domestic robot in the form of a

robotic dog with intelligent features and autonomy.

Mattel created an assortment of AI-enabled toys that "understand" conversations, give intelligent responses, and learn.

Oil and gas

Oil and gas companies have used artificial intelligence tools to automate functions, foresee equipment issues, and increase oil and gas output.

Mathematics

AI tools have been used to translate mathematical proofs into formal proofs in order to automatically verify them.

Automated theorem proving

Computational geometry

AI-assisted mathematical discovery

FunSearch, developed by Google DeepMind, combines a large language model with automated evaluation to search for programs that can help solve mathematical problems. It has been applied to extremal combinatorics, including the cap set problem, and to online bin packing.

AI-assisted algorithm discovery

AI systems have also been used to search for improved algorithms and computational routines. AlphaTensor, AlphaDev, and AlphaEvolve have been applied to matrix multiplication algorithms, sorting

routines, and broader scientific and algorithmic discovery.

Military

Various countries are deploying AI military applications. Research is targeting intelligence collection and analysis, logistics, cyber operations, information operations, and semiautonomous and autonomous vehicles.

AI has been used in military operations in Iraq, Syria, Israel and Ukraine.

Robotics

Artificial intelligence is used in robotics to help robots perceive their surroundings, plan actions, navigate, manipulate objects, and adapt to changing environments. AI systems are used to connect sensor data with physical action, especially in systems operating outside controlled factory settings.

Internet and e-commerce

Procurement and Sourcing

AI has been used in B2B sourcing and procurement. Academic literature and industry reports point out uses of AI for supplier evaluation and selection, predictive demand forecasting, automated contract and invoice processing, and supply-chain risk assessment. Machine learning, natural language processing, and robotic process automation are commonly used in procurement.

Web feeds and posts

Machine learning has been used for recommendation systems in determining which posts should show up in social media feeds. Various types of social media analysis also make use of machine learning and there is research into its use for (semi-)automated

tagging/enhancement/correction of online misinformation and related filter bubbles.

AI has been used to customize shopping options and personalize offers. Online gambling companies have used AI for targeting gamblers.

Virtual assistants and search

Intelligent personal assistants use AI to attempt to respond to natural language requests. Siri, released in 2010 for Apple smartphones, popularized the concept.

Bing Chat has used artificial intelligence as part of its search engine.

Spam filtering

Machine learning can be used to combat spam, scams, and phishing. It can scrutinize the contents of spam and phishing attacks to attempt to identify malicious elements. Some models built via machine learning algorithms have over 90% accuracy in distinguishing between spam and legitimate emails. These models can be refined using new data and evolving spam tactics. Machine learning also analyzes traits such as sender behavior, email header information, and attachment types, potentially enhancing spam detection.

Facial recognition and image labeling

AI has been used in facial recognition systems. Some examples are Apple's Face ID and Android's Face Unlock, which are used to secure mobile devices.

China has used facial recognition and artificial intelligence technology in Xinjiang. In 2017, reporters visiting the region found surveillance cameras installed every hundred meters or so in several cities, as well as facial recognition checkpoints at areas like gas stations, shopping centers, and mosque entrances. Human rights groups have criticized the Chinese government for using artificial intelligence facial recognition technology for use in political suppression.

The Netherlands has deployed facial recognition and artificial intelligence technology since

2016. The database of the Dutch police currently contains over 2.2 million pictures of 1.3 million Dutch citizens. This accounts for about 8% of the population. In The Netherlands, face recognition is not used by the police on municipal CCTV.

Image labeling has been used by Google Image Labeler to detect products in photos and to allow people to search based on a photo. Image labeling has also been demonstrated to generate speech to describe images to blind people.

Scientific research

Evidence of general impacts

In April 2024, the Scientific Advice Mechanism to the European Commission published advice including a comprehensive evidence review of the opportunities and challenges posed by artificial intelligence in scientific research.

As benefits, the evidence review highlighted:

- its role in accelerating research and innovation

- its capacity to automate workflows

- enhancing dissemination of scientific work

As challenges:

- limitations and risks around transparency, reproducibility and interpretability

- poor performance (inaccuracy)

- risk of harm through misuse or unintended use

- societal concerns including the spread of misinformation and increasing inequalities

Archaeology, history and imaging of sites

Machine learning can help to restore and attribute ancient texts. It can help to index texts for

example to enable better and easier searching and classification of fragments.

Artificial intelligence can also be used to investigate genomes to uncover genetic history, such as interbreeding between archaic and modern humans by which for example the past existence of a ghost population, not Neanderthal or Denisovan, was inferred.

It can also be used for "non-invasive and non-destructive access to internal structures of archaeological remains".

Physics

A deep learning system was reported to learn intuitive physics from visual data (of virtual 3D environments) based on an unpublished approach inspired by studies of visual cognition in infants. Other researchers have developed a machine learning algorithm that could discover sets of basic variables of various physical systems and predict the systems' future dynamics from video recordings of their behavior. In the future, it may be possible that such can be used to automate the discovery of physical laws of complex systems.

Materials science

In November 2023, researchers at Google DeepMind and Lawrence Berkeley National Laboratory announced that the AI system GNoME had documented over 2 million new materials. GNoME uses deep learning techniques to examine potential material structures, and identify stable inorganic crystal structures. The system's predictions were validated through autonomous robotic experiments, with a success rate of 71%. The data of newly discovered materials is publicly available through the Materials Project database.

Reverse engineering

Machine learning is used in diverse types of reverse engineering. For example, machine

learning has been used to reverse engineer a composite material part, enabling unauthorized production of high quality parts, and for quickly understanding the behavior of malware. It can be used to reverse engineer artificial intelligence models. It can also design components by engaging in a type of reverse engineering of not-yet existent virtual components such as inverse molecular design for particular desired functionality or protein design for pre-specified functional sites. Biological network reverse engineering could model interactions in a human understandable way, e.g. bas on time series data of gene expression levels.

Astronomy, space activities and ufology

Artificial intelligence is used in astronomy to analyze increasing amounts of available data and applications, mainly for "classification, regression, clustering, forecasting, generation, discovery, and the development of new scientific insights" for example for discovering exoplanets, forecasting solar activity, and distinguishing between signals and instrumental effects in gravitational wave astronomy. It could also be used for activities in space such as space exploration, including analysis of data from space missions, real-time science decisions of spacecraft, space debris avoidance, and more autonomous operation.

In the search for extraterrestrial intelligence (SETI), machine learning has been used in attempts to identify artificially generated electromagnetic waves in available data – such as real-time observations – and other technosignatures, e.g. via anomaly detection. In ufology, the SkyCAM-5 project headed by Prof. Hakan Kayal and the Galileo Project headed by Avi Loeb use machine learning to attempt to detect and classify types of UFOs. The Galileo Project also seeks to detect two further types of potential extraterrestrial technological signatures with the use of AI: 'Oumuamua-like interstellar objects, and non-manmade artificial satellites.

Machine learning can also be used to produce datasets of spectral signatures of molecules that may be involved in the atmospheric production or consumption of particular chemicals – such as phosphine possibly detected on Venus – which could prevent miss assignments and, if

accuracy is improved, be used in future detections and identifications of molecules on other planets.

Chemistry and biology

There is research about which types of computer-aided chemistry would benefit from machine learning. A deep learning AI-based process has been developed that uses genome databases to design novel proteins based on evolutionary algorithms.

Machine learning has also been used for protein design with pre-specified functional sites, predicting molecular properties, and exploring large chemical/reaction spaces.

Using drug discovery AI algorithms, researchers generated 40,000 potential chemical weapon candidates, helping in the regulation of such chemicals to prevent synthesizing them for real harm.

There are various types of applications for machine learning in decoding human biology, such as helping to map gene expression patterns to functional activation patterns or identifying functional DNA motifs. It is widely used in genetic research.

There also is some use of machine learning in synthetic biology, disease biology, nanotechnology (e.g. nanostructured materials and bionanotechnology), and materials science.

Security and surveillance

Cyber security

Cyber security companies are adopting neural networks, machine learning, and natural language processing to improve their systems.

Applications of AI in cyber security include:

Network protection: Machine learning improves intrusion detection systems by broadening

the search beyond previously identified threats.

Endpoint protection: Attacks such as ransomware can be thwarted by learning typical malware behaviors.

AI-related cyber security application cases vary in both benefit and complexity. Security features such as Security Orchestration, Automation, and Response (SOAR) and Extended Endpoint Detection and Response (XDR) offer significant benefits for businesses, but require significant integration and adaptation efforts.

Application security: can help counterattacks such as server-side request forgery, SQL injection, cross-site scripting, and distributed denial-of-service.

AI technology can also be utilized to improve system security and safeguard our privacy. Randrianasolo (2012) suggested a security system based on artificial intelligence that can recognize intrusions and adapt to perform better. In order to improve cloud computing security, Sahil (2015) created a user profile system for the cloud environment with AI techniques.

Suspect user behavior: Machine learning can identify fraud or compromised applications as they occur.

Transportation and logistics

Aviation

The use of artificial intelligence is increasingly observed in the aviation industry for predictive maintenance. This has been helpful in reducing delays. Companies such as Boeing and Airbus are using AI to optimize repair procedures while maintaining operational performance.

Automotive and public transit

Transportation's complexity means that in most cases, training an AI in a real-world driving environment is impractical, and is achieved through simulator-based testing. AI-based

systems control functions such as braking, lane changing, collision prevention, navigation and mapping.

Some autonomous vehicles do not allow human drivers (they have no steering wheels or pedals).

There are prototypes of autonomous automotive public transport vehicles such as autonomous rail transport in operation, electric mini-buses, and autonomous delivery vehicles, including delivery robots.

Autonomous trucks are in the testing phase. The UK government passed legislation to begin testing of autonomous truck platoons in 2018. A group of autonomous trucks follow closely behind each other. German corporation Daimler is testing its Freightliner Inspiration.

AI has been used to optimize traffic management, which can reduce wait times, energy use, and emissions.

Military

Aircraft simulators use AI for training aviators. Flight conditions can be simulated that allow pilots to make mistakes without risking themselves or expensive aircraft. Air combat can also be simulated.

AI can also be used to operate planes analogously to their control of ground vehicles. Autonomous drones can fly independently or in swarms.

AOD uses the Interactive Fault Diagnosis and Isolation System, or IFDIS, which is a rule-based expert system using information from TF-30 documents and expert advice from mechanics that work on the TF-30. This system was designed to be used for the development of the TF-30 for the F-111C. The system replaced specialized workers. The system allowed regular workers to communicate with the system and avoid mistakes, miscalculations, or having to speak to one of the specialized workers.

Speech recognition allows traffic controllers to give verbal directions to drones.

Artificial intelligence supported design of aircraft, or AIDA, is used to help designers in the process of creating conceptual designs of aircraft. This program allows the designers to focus more on the design itself and less on the design process. The software also allows the user to focus less on the software tools. The AIDA uses rule-based systems to compute its data. This is a diagram of the arrangement of the AIDA modules. Although simple, the program is proving effective.

NASA

In 2003 a Dryden Flight Research Center project created software that could enable a damaged aircraft to continue flight until a safe landing can be achieved. The software compensated for damaged components by relying on the remaining undamaged components.

The 2016 Intelligent Autopilot System combined apprenticeship learning and behavioral cloning whereby the autopilot observed low-level actions required to maneuver the airplane and high-level strategy used to apply those actions.

Maritime

Neural networks are used by situational awareness systems in ships and boats. There also are autonomous boats.

See also

Applications of artificial intelligence to legal informatics

Applications of deep learning

Applications of machine learning

Artificial intelligence and elections

Collective intelligence § Applications

List of artificial intelligence projects

List of datasets for machine-learning research

Open data

Progress in artificial intelligence

Timeline of computing 2020–present

Footnotes

Further reading

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